



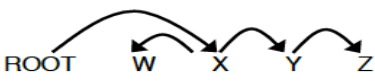
Indian Institute of Technology Bhubaneswar
School of Electrical and Computer Sciences
Mid-Semester (Spring) Examination – 2026

Subject Name: Natural Language Processing

Subject Code: CS6L027

1. Multiple choice questions [There could be multiple correct options, no partial marking]

- 1.1. b. 4
1.2. c. I said she stole someone else's money.
1.3. b. 5

- 1.4. c. 
1.5. b. 0100 d. 0101
-

2.

- 2.1. The phrase “that no one talks about” could either be referring to the cost, “No one talks about the awful cost of getting a PhD” or to the PhD, “Getting a PhD that no one talks about is awfully costly”.
- 2.2. The problem addressed by a PoS tagger is to assign part-of-speech tags (i.e., grammatical roles) to words within a given context (e.g., a sentence or text). This task is not trivial because of lexical ambiguity (words can have multiple grammatical roles, e.g. can/N can/V) and out-of-vocabulary forms (i.e. unknown words).
Lexical ambiguity is not trivial to handle because it leads to an exponential number of possible solution w.r.t the sentence length.

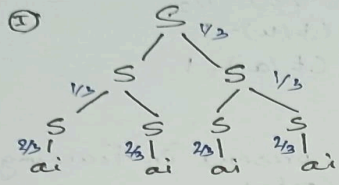
Unknown words are not trivial because we must decide how to handle them, a process that often involves high-level linguistic features.

3.

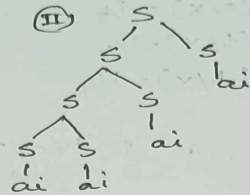
③

5 parse trees

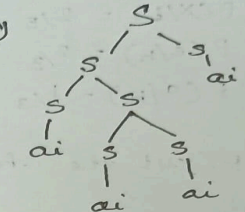
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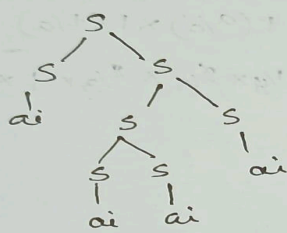
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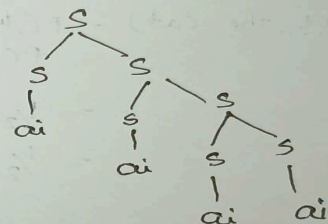
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④



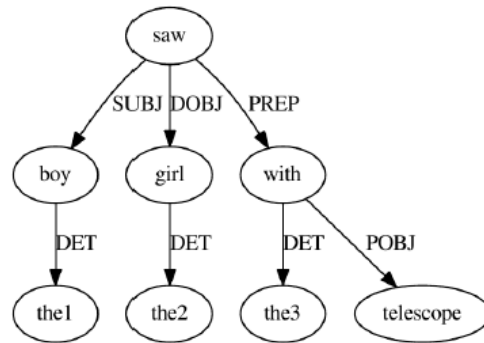
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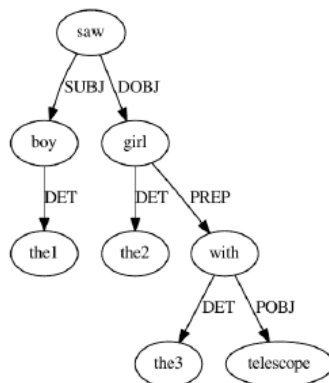
Probability of Each Parse tree = $\frac{2^4}{3^7}$

Probability of 'ai ai ai ai' = $5 \times \frac{2^4}{3^7}$

4.



The boy uses the telescope to see the girl.



The boy saw the girl who had the telescope.

5.

5.1.

There are 12 different bigrams (denoting here the whitespace with 'X' to better see it):
Xc, Xh, Xt, at, ca, cu, eX, ha, he, tX, th, ut,

Here are the counts

Xc, 2; Xh, 1; Xt, 1; at, 2; ca, 1; cu, 1; eX, 2; ha, 1; he, 2; tX, 2; th, 2; ut, 1

5.2.

5.2

Let's assume, space character is denoted by 'X'

$$P(h/t) = 2/5$$

$$P(X/e) = 1$$

$$P(a/c) = 1/2$$

$$P(X/t) = 2/5$$

$$P(c/x) = 1/2$$

$$P(u/c) = 1/2$$

$$P(e/h) = 2/3$$

$$P(t/x) = 1/4$$

$$P(t/u) = 1$$

$$P(a/h) = 1/3$$

$$P(h/x) = 1/4$$

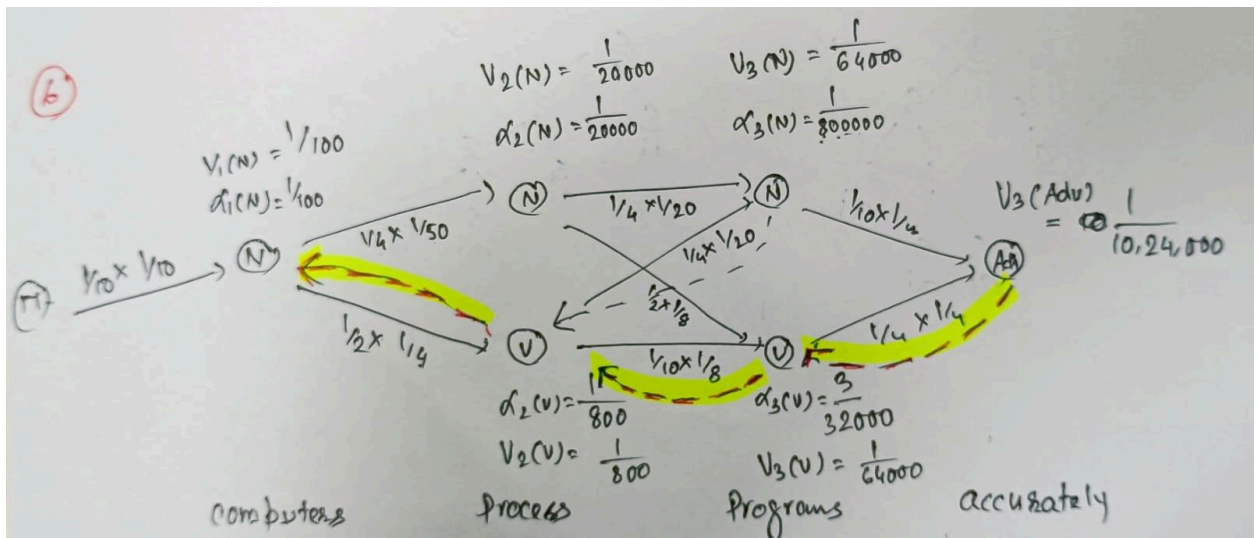
$$P(t/a) = 1$$

$P(\text{cut the chat}) = 0$, as $P(h/c) = 0$
 $\hookrightarrow$ not present in training corpus.

$$P(\text{cut the cat}) = P(u/c) \times P(t/u) \times P(X/t) \times P(t/x) \times P(h/t) \times P(a/h) \times P(X/e) \times P(c/x) \times P(a/c) \times P(t/a)$$

$$= 1/2 \times 1 \times 2/5 \times 1/4 \times 2/5 \times 2/3 \times 1 \times 1/2 \times 1/2 \times 1$$

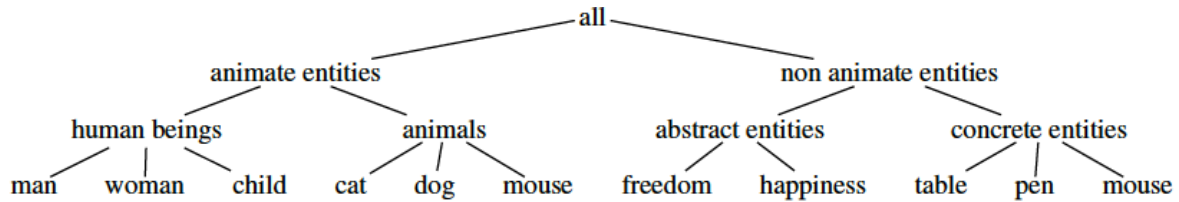
$$= 1/300$$



6.

So the most probable sequence is N V V Adv

7. Consider the following ontology providing a semantic structuring for a (small) set of nouns:



- 7.1. What is the semantic relation that has been used to build the ontology? (1)

Solution: *"is a" relation: hypernymy*

- 7.2. The word "mouse" appears twice in the ontology. What does this mean? What specific problems does it raise when the ontology is used? How could such problems be solved? (Just provide a sketch of explanation.) (2)

Solution: *semantic ambiguity (two different meanings, polysemy, homonymy(homography)). Word Sense Disambiguation (WSD) using information from context (e.g., cohesion).*

- 7.3. Consider the following short text:

Cats are fighting dogs. There are plenty of pens on the table.

What pre-processing should be performed on this text to make it suitable for the use of the available ontology? (2)

Solution: *identify surface forms (tokenization), maybe stopwords filtering, PoS tagging for nouns, lemmatization (or stemming).*

8.

8.1.

line 7: "width := 2 to n" --> "width := 3 to n"

line 10: "mid := start+1 to end-1" --> "mid := start+2 to end-1"

line 11: "for (X->Y Z) in Rules" --> "for (X->Y a Z) in Rules"

line 12: "if Chart[start,mid,Y] and Chart[mid,end,Z]" --> "if Chart[start,mid-1,Y] and Input[mid]=a and Chart[mid,end,Z]"

8.2. *Each ternary rule $X \rightarrow Y a Z$ should be replaced with the rules*

$X \rightarrow Y Z'$

$Z' \rightarrow A Z$

$A \rightarrow a$

where the names Z' and A are chosen to be new nonterminals not already in the grammar.